

Lecture 11: Properties of OLS in multiple regression

Economics 326 — Introduction to Econometrics II

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Multiple regression and OLS

- Consider the multiple regression model with k regressors:

$$Y_i = \beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \dots + \beta_k X_{k,i} + U_i.$$

- Let $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ be the OLS estimators: if

$$\hat{U}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1,i} - \hat{\beta}_2 X_{2,i} - \dots - \hat{\beta}_k X_{k,i},$$

then

$$\sum_{i=1}^n \hat{U}_i = \sum_{i=1}^n X_{1,i} \hat{U}_i = \dots = \sum_{i=1}^n X_{k,i} \hat{U}_i = 0.$$

Multiple regression and OLS

- As in Lecture 9, we can write $\hat{\beta}_1$ as

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n \tilde{X}_{1,i} Y_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2}, \text{ where}$$

– $\tilde{X}_{1,i}$ are the fitted OLS residuals: $\tilde{X}_{1,i} = X_{1,i} - \hat{\gamma}_0 - \hat{\gamma}_2 X_{2,i} - \dots - \hat{\gamma}_k X_{k,i}$.

– $\hat{\gamma}_0, \hat{\gamma}_2, \dots, \hat{\gamma}_k$ are the OLS coefficients: $\sum_{i=1}^n \tilde{X}_{1,i} = \sum_{i=1}^n \tilde{X}_{1,i} X_{2,i} = \dots = \sum_{i=1}^n \tilde{X}_{1,i} X_{k,i} = 0$.

- Similarly, we can write $\hat{\beta}_2$ as

$$\hat{\beta}_2 = \frac{\sum_{i=1}^n \tilde{X}_{2,i} Y_i}{\sum_{i=1}^n \tilde{X}_{2,i}^2}, \text{ where}$$

– $\tilde{X}_{2,i}$ are the fitted OLS residuals: $\tilde{X}_{2,i} = X_{2,i} - \hat{\delta}_0 - \hat{\delta}_1 X_{1,i} - \hat{\delta}_3 X_{3,i} - \dots - \hat{\delta}_k X_{k,i}$.

– $\hat{\delta}_0, \hat{\delta}_1, \hat{\delta}_3, \dots, \hat{\delta}_k$ are the OLS coefficients: $\sum_{i=1}^n \tilde{X}_{2,i} = \sum_{i=1}^n \tilde{X}_{2,i} X_{1,i} = \sum_{i=1}^n \tilde{X}_{2,i} X_{3,i} = \dots = \sum_{i=1}^n \tilde{X}_{2,i} X_{k,i} = 0$.

The OLS estimators are linear

- Consider $\hat{\beta}_1$:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n \tilde{X}_{1,i} Y_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2} = \sum_{i=1}^n \frac{\tilde{X}_{1,i}}{\sum_{l=1}^n \tilde{X}_{1,l}^2} Y_i = \sum_{i=1}^n w_{1,i} Y_i,$$

where

$$w_{1,i} = \frac{\tilde{X}_{1,i}}{\sum_{l=1}^n \tilde{X}_{1,l}^2}.$$

- Recall that $\tilde{X}_1$ are the residuals from a regression of X_1 on $X_2, \dots, X_k$ and a constant, and therefore $w_{1,i}$ depends only on $\mathbf{X}$.

Unbiasedness

- Suppose that
 - $Y_i = \beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \dots + \beta_k X_{k,i} + U_i$.
 - Conditional on $\mathbf{X}$** , $E[U_i | \mathbf{X}] = 0$ for all i 's.
 - Conditioning on $\mathbf{X}$ means that we condition on all regressors for all observations: $\mathbf{X} = \{(X_{1,i}, X_{2,i}, \dots, X_{k,i}) : i = 1, \dots, n\}$.
- Under the above assumptions, conditional on $\mathbf{X}$:

$$E[\hat{\beta}_0 | \mathbf{X}] = \beta_0,$$

$$E[\hat{\beta}_1 | \mathbf{X}] = \beta_1,$$

⋮

$$E[\hat{\beta}_k | \mathbf{X}] = \beta_k.$$

Proof of unbiasedness

- Substituting Y_i and expanding:

$$\begin{aligned} \hat{\beta}_1 &= \frac{\sum_{i=1}^n \tilde{X}_{1,i} Y_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2} = \frac{\sum_{i=1}^n \tilde{X}_{1,i} (\beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \dots + \beta_k X_{k,i} + U_i)}{\sum_{i=1}^n \tilde{X}_{1,i}^2} \\ &= \beta_0 \frac{\sum_{i=1}^n \tilde{X}_{1,i}}{\sum_{i=1}^n \tilde{X}_{1,i}^2} + \beta_1 \frac{\sum_{i=1}^n \tilde{X}_{1,i} X_{1,i}}{\sum_{i=1}^n \tilde{X}_{1,i}^2} + \beta_2 \frac{\sum_{i=1}^n \tilde{X}_{1,i} X_{2,i}}{\sum_{i=1}^n \tilde{X}_{1,i}^2} \\ &\quad + \dots + \beta_k \frac{\sum_{i=1}^n \tilde{X}_{1,i} X_{k,i}}{\sum_{i=1}^n \tilde{X}_{1,i}^2} + \frac{\sum_{i=1}^n \tilde{X}_{1,i} U_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2}. \end{aligned}$$

- Using the partitioned regression results from Lecture 9:

$$\begin{aligned} \sum_{i=1}^n \tilde{X}_{1,i} &= \sum_{i=1}^n \tilde{X}_{1,i} X_{2,i} = \dots = \sum_{i=1}^n \tilde{X}_{1,i} X_{k,i} = 0, \\ \sum_{i=1}^n \tilde{X}_{1,i} X_{1,i} &= \sum_{i=1}^n \tilde{X}_{1,i}^2. \end{aligned}$$

- Therefore,

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^n \tilde{X}_{1,i} U_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2}.$$

Proof of unbiasedness

- We have

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^n \tilde{X}_{1,i} U_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2}.$$

- **Conditional on $\mathbf{X}$,**

$$E[U_i | \mathbf{X}] = 0.$$

- Therefore, **conditional on $\mathbf{X}$,**

$$\begin{aligned} E[\hat{\beta}_1 | \mathbf{X}] &= E\left[\beta_1 + \frac{\sum_{i=1}^n \tilde{X}_{1,i} U_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2} \mid \mathbf{X}\right] \\ &= \beta_1 + \frac{\sum_{i=1}^n \tilde{X}_{1,i} E[U_i | \mathbf{X}]}{\sum_{i=1}^n \tilde{X}_{1,i}^2} \\ &= \beta_1. \end{aligned}$$

Conditional variance of the OLS estimators

- Suppose that:
 1. $Y_i = \beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \dots + \beta_k X_{k,i} + U_i$.
 2. **Conditional on $\mathbf{X}$,** $E[U_i | \mathbf{X}] = 0$ for all i 's.
 3. **Conditional on $\mathbf{X}$,** $E[U_i^2 | \mathbf{X}] = \sigma^2$ for all i 's.
 4. **Conditional on $\mathbf{X}$,** $E[U_i U_j | \mathbf{X}] = 0$ for all $i \neq j$.
- Denote by $SSR_1 = \sum_{i=1}^n \tilde{X}_{1,i}^2$ the residual sum-of-squares from regressing X_1 on a constant and the other regressors. The conditional variance of $\hat{\beta}_1$ given $\mathbf{X}$ is

$$\text{Var}(\hat{\beta}_1 | \mathbf{X}) = \frac{\sigma^2}{\sum_{i=1}^n \tilde{X}_{1,i}^2} = \frac{\sigma^2}{SSR_1}.$$

- **Gauss-Markov Theorem:** Under Assumptions 1–4, the OLS estimators are **BLUE**.

Derivation of the conditional variance

- We have $\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^n \tilde{X}_{1,i} U_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2}$.

- Conditional on $\mathbf{X}$,

$$\begin{aligned}
\text{Var}(\hat{\beta}_1 | \mathbf{X}) &= \text{E} \left[\left(\hat{\beta}_1 - \text{E}[\hat{\beta}_1 | \mathbf{X}] \right)^2 | \mathbf{X} \right] \\
&= \text{E} \left[\left(\frac{\sum_{i=1}^n \tilde{X}_{1,i} U_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2} \right)^2 | \mathbf{X} \right] \\
&= \frac{1}{\left(\sum_{i=1}^n \tilde{X}_{1,i}^2 \right)^2} \text{E} \left[\left(\sum_{i=1}^n \tilde{X}_{1,i} U_i \right)^2 | \mathbf{X} \right] \\
&= \frac{1}{\left(\sum_{i=1}^n \tilde{X}_{1,i}^2 \right)^2} \left(\sum_{i=1}^n \tilde{X}_{1,i}^2 \sigma^2 + \sum_{i \neq j} \tilde{X}_{1,i} \tilde{X}_{1,j} \cdot 0 \right) \\
&= \frac{\sigma^2 \sum_{i=1}^n \tilde{X}_{1,i}^2}{\left(\sum_{i=1}^n \tilde{X}_{1,i}^2 \right)^2} = \frac{\sigma^2}{\sum_{i=1}^n \tilde{X}_{1,i}^2} = \frac{\sigma^2}{SSR_1}.
\end{aligned}$$

Conditional covariance of the OLS estimators

- Consider $\hat{\beta}_1$ and $\hat{\beta}_2$:

$$\begin{aligned}
\hat{\beta}_1 &= \beta_1 + \frac{\sum_{i=1}^n \tilde{X}_{1,i} U_i}{SSR_1}, \\
\hat{\beta}_2 &= \beta_2 + \frac{\sum_{i=1}^n \tilde{X}_{2,i} U_i}{SSR_2},
\end{aligned}$$

where $SSR_2 = \sum_{i=1}^n \tilde{X}_{2,i}^2$ is the residual sum-of-squares from regressing X_2 on a constant and $X_1, X_3, \dots, X_k$.

- We will show that given Assumptions 1–4, **conditional on $\mathbf{X}$** :

$$\text{Cov}(\hat{\beta}_1, \hat{\beta}_2 | \mathbf{X}) = \sigma^2 \frac{\sum_{i=1}^n \tilde{X}_{1,i} \tilde{X}_{2,i}}{SSR_1 \cdot SSR_2}$$

Conditional covariance of the OLS estimators

Conditional on $\mathbf{X}$,

$$\begin{aligned}
\text{Cov}(\hat{\beta}_1, \hat{\beta}_2 | \mathbf{X}) &= \text{E} \left[\left(\hat{\beta}_1 - \text{E}[\hat{\beta}_1 | \mathbf{X}] \right) \left(\hat{\beta}_2 - \text{E}[\hat{\beta}_2 | \mathbf{X}] \right) | \mathbf{X} \right] \\
&= \frac{1}{SSR_1 \cdot SSR_2} \text{E} \left[\left(\sum_{i=1}^n \tilde{X}_{1,i} U_i \right) \left(\sum_{i=1}^n \tilde{X}_{2,i} U_i \right) | \mathbf{X} \right] \\
&= \frac{1}{SSR_1 \cdot SSR_2} \left(\sum_{i=1}^n \tilde{X}_{1,i} \tilde{X}_{2,i} \sigma^2 + \sum_{i \neq j} \tilde{X}_{1,i} \tilde{X}_{2,j} \cdot 0 \right) \\
&= \sigma^2 \frac{\sum_{i=1}^n \tilde{X}_{1,i} \tilde{X}_{2,i}}{SSR_1 \cdot SSR_2}.
\end{aligned}$$

Normality of the OLS estimators

- In addition to Assumptions 1–4, assume that **conditional on $\mathbf{X}$** , U_i 's are jointly normally distributed.

- $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ are linear estimators:

$$\hat{\beta}_j = \sum_{i=1}^n w_{j,i} Y_i = \beta_j + \sum_{i=1}^n w_{j,i} U_i,$$

where

$$w_{j,i} = \frac{\tilde{X}_{j,i}}{\sum_{l=1}^n \tilde{X}_{j,l}^2},$$

and $\tilde{X}_{j,i}$ are the residuals from the regression of X_j on the rest of the regressors.

- It follows that $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ are jointly normally distributed (conditional on $\mathbf{X}$).

Inclusion of irrelevant regressors: No bias

- Suppose that the true model is $Y_i = \beta_0 + \beta_1 X_{1,i} + U_i$.
- We could estimate β_1 by

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_{1,i} - \bar{X}_1) Y_i}{\sum_{i=1}^n (X_{1,i} - \bar{X}_1)^2}.$$

- Suppose that instead we regress Y on a constant, X_1 , and $k-1$ additional regressors $X_2, \dots, X_k$, i.e., we estimate β_1 by

$$\tilde{\beta}_1 = \frac{\sum_{i=1}^n \tilde{X}_{1,i} Y_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2}.$$

- We have

$$\begin{aligned} \tilde{\beta}_1 &= \frac{\sum_{i=1}^n \tilde{X}_{1,i} (\beta_0 + \beta_1 X_{1,i} + U_i)}{\sum_{i=1}^n \tilde{X}_{1,i}^2} \\ &= \beta_1 + \frac{\sum_{i=1}^n \tilde{X}_{1,i} U_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2}. \end{aligned}$$

- Since conditional on $\mathbf{X}$, $E[U_i | \mathbf{X}] = 0$, $\tilde{\beta}_1$ is **unbiased**!

Inclusion of irrelevant regressors: Variance inflation

- When $Y_i = \beta_0 + \beta_1 X_{1,i} + U_i$,

$$\begin{aligned} \hat{\beta}_1 &= \frac{\sum_{i=1}^n (X_{1,i} - \bar{X}_1) Y_i}{\sum_{i=1}^n (X_{1,i} - \bar{X}_1)^2} \quad \text{and} \\ \tilde{\beta}_1 &= \frac{\sum_{i=1}^n \tilde{X}_{1,i} Y_i}{\sum_{i=1}^n \tilde{X}_{1,i}^2} \end{aligned}$$

are both unbiased.

- Their variances, conditional on $\mathbf{X}$:

$$\text{Var}(\hat{\beta}_1 | \mathbf{X}) = \frac{\sigma^2}{\sum_{i=1}^n (X_{1,i} - \bar{X}_1)^2} \quad \text{and}$$

$$\text{Var}(\tilde{\beta}_1 | \mathbf{X}) = \frac{\sigma^2}{SSR_1}.$$

- In the short regression, X_1 is regressed on a constant only, so $SSR_1 = \sum_{i=1}^n (X_{1,i} - \bar{X}_1)^2$.
- In the long regression, $X_2, \dots, X_k$ are added. From Lecture 10, this cannot increase SSR_1 , so

$$\text{Var}(\hat{\beta}_1 | \mathbf{X}) \leq \text{Var}(\tilde{\beta}_1 | \mathbf{X}).$$

- Including irrelevant regressors inflates the variance of $\hat{\beta}_1$.

Variance and the number of regressors

- Recall the variance formula:

$$\text{Var}(\hat{\beta}_1 | \mathbf{X}) = \frac{\sigma^2}{SSR_1}.$$

- When we add a new regressor to the model, two quantities may change:
 1. σ^2 (the error variance): if the new regressor genuinely explains variation in Y , including it in the model moves that variation from U_i into the explained part, reducing $\text{Var}(U_i | \mathbf{X}) = \sigma^2$
 2. SSR_1 (the variation in X_1 net of the other regressors): can only decrease or stay the same; stays the same only when the new regressor is uncorrelated with X_1

Case A: Irrelevant regressor

- Suppose the new regressor does not affect Y (its population coefficient is zero).
- σ^2 is unchanged, because the error variance is determined by the true data-generating process.
- SSR_1 decreases if the new regressor is correlated with X_1 .
- Net effect: $\text{Var}(\hat{\beta}_1 | \mathbf{X})$ **increases**.
- **Conclusion:** do not include irrelevant regressors.

Case B: Relevant regressor, uncorrelated with X_1

- Suppose the new regressor affects Y but is uncorrelated with X_1 after controlling for the other regressors.
- SSR_1 is approximately unchanged: the new regressor has no additional predictive power for X_1 beyond the existing regressors, so adding it to the auxiliary regression has a negligible effect on SSR_1 in large samples.
- σ^2 decreases, because the new regressor explains part of the variation in Y .
- Net effect: $\text{Var}(\hat{\beta}_1 | \mathbf{X})$ **decreases**.
- **Conclusion:** always include such regressors.

Case C: Relevant regressor, correlated with X_1

- Two opposing forces:
 - σ^2 decreases (the new regressor explains variation in Y)
 - SSR_1 decreases (the new regressor is correlated with X_1)
- Net effect on $\text{Var}(\hat{\beta}_1 | \mathbf{X})$ is **ambiguous**: both the numerator and the denominator of σ^2/SSR_1 shrink, so the ratio could go either way.
- However, omitting a relevant regressor introduces **omitted variable bias** (see Lecture 9).
- Conclusion**: always include relevant regressors, even if the variance of $\hat{\beta}_1$ may increase. An unbiased estimator with larger variance is preferable to a biased one.

Estimation of variances and covariances

- In $Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1,i} + \hat{\beta}_2 X_{2,i} + \dots + \hat{\beta}_k X_{k,i} + \hat{U}_i$,

$$\text{Var}(\hat{\beta}_1 | \mathbf{X}) = \frac{\sigma^2}{SSR_1},$$
$$\text{Cov}(\hat{\beta}_1, \hat{\beta}_2 | \mathbf{X}) = \sigma^2 \frac{\sum_{i=1}^n \tilde{X}_{1,i} \tilde{X}_{2,i}}{SSR_1 \cdot SSR_2}.$$

- Variances and covariances can be estimated by replacing σ^2 with

$$s^2 = \frac{1}{n-k-1} \sum_{i=1}^n \hat{U}_i^2.$$

- Estimated variance and covariance:

$$\widehat{\text{Var}}(\hat{\beta}_1) = \frac{s^2}{SSR_1},$$
$$\widehat{\text{Cov}}(\hat{\beta}_1, \hat{\beta}_2) = s^2 \frac{\sum_{i=1}^n \tilde{X}_{1,i} \tilde{X}_{2,i}}{SSR_1 \cdot SSR_2}.$$

Standard errors in terms of R-squared

- Auxiliary R^2** : Let R_1^2 be the R-squared from regressing X_1 on a constant and $X_2, \dots, X_k$. By definition of R-squared,

$$SSR_1 = SST_1(1 - R_1^2), \quad \text{where } SST_1 = \sum_{i=1}^n (X_{1,i} - \bar{X}_1)^2.$$

- Adjusted R^2** : From $s^2 = s_Y^2(1 - \bar{R}^2)$, where $s_Y^2 = SST/(n-1)$ is the sample variance of Y ,

$$\text{se}(\hat{\beta}_1) = \sqrt{\frac{s^2}{SSR_1}} = \sqrt{\frac{s_Y^2(1 - \bar{R}^2)}{SST_1(1 - R_1^2)}}.$$

- Define $s_{X_1} = \sqrt{SST_1/(n-1)}$. Then

$$\text{se}(\hat{\beta}_1) = \frac{s_Y}{s_{X_1}} \cdot \frac{1}{\sqrt{n-1}} \cdot \sqrt{\frac{1 - \bar{R}^2}{1 - R_1^2}}.$$

Three factors behind standard errors

- The SE formula has three interpretable factors:

$$\text{se}(\hat{\beta}_1) = \underbrace{\frac{s_Y}{s_{X_1}}}_{\text{scaling}} \cdot \underbrace{\frac{1}{\sqrt{n-1}}}_{\text{sample size}} \cdot \underbrace{\sqrt{\frac{1-\bar{R}^2}{1-R_1^2}}}_{\text{fit vs. collinearity}}$$

1. s_Y/s_{X_1} — scaling: the ratio of sample standard deviations of Y and X_1
2. $1/\sqrt{n-1}$ — sample size effect; more data reduces SE
3. $\sqrt{(1-\bar{R}^2)/(1-R_1^2)}$ — fit vs. multicollinearity trade-off:
 - $(1-\bar{R}^2)$: unexplained variation in Y (adjusted for degrees of freedom); higher $\bar{R}^2$ reduces SE
 - $(1-R_1^2)$: unique variation in X_1 ; more collinearity (higher R_1^2) inflates SE

Connection to Cases A, B, C

- The SE formula clarifies the three cases:
 - **Case A** (irrelevant regressor): $\bar{R}^2$ decreases, R_1^2 may increase $\implies$ SE increases
 - **Case B** (relevant, uncorrelated with X_1): $\bar{R}^2$ increases, $R_1^2 \approx$ unchanged $\implies$ SE decreases
 - **Case C** (relevant, correlated with X_1): $\bar{R}^2$ increases but R_1^2 also increases $\implies$ ambiguous
- Remark.** An equivalent expression uses the unadjusted R^2 :

$$\text{se}(\hat{\beta}_1) = \frac{s_Y}{s_{X_1}} \cdot \frac{1}{\sqrt{n-k-1}} \cdot \sqrt{\frac{1-R^2}{1-R_1^2}}.$$

This follows from $s^2 = SST(1-R^2)/(n-k-1)$, which keeps R^2 and the degrees-of-freedom correction separate.